HOPE LAB / METHOD REFERENCE

DPPH Radical Scavenging Assay for Antioxidant Activity

A source-based guide to assay design, blank correction, calculations and interpretation.

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Purpose and principle

The DPPH radical-scavenging assay estimates the ability of a sample to reduce the stable purple DPPH radical. The decrease in absorbance is commonly monitored in the visible range around 515-517 nm. Reaction kinetics vary by antioxidant and matrix, so a single universal endpoint should not be assumed.

Study-design workflow (non-operational)

- Define the test material and the scientific claim. The DPPH assay measures a chemical radical-scavenging response in vitro; it does not establish in-vivo antioxidant efficacy or health benefit.
- Select a validated, institutionally approved assay variant that is compatible with sample solubility and the available instrument. Follow the local chemical risk assessment and current reagent safety documentation.
- Plan a concentration-response series and include a solvent/reagent control, a sample-colour or turbidity blank where necessary, and a recognized positive antioxidant reference.
- Use the same assay variant, solvent system, reaction endpoint and measurement settings across the comparison set. Record any matrix-specific modifications.
- Correct sample absorbance for intrinsic colour or turbidity before calculating scavenging response.
- Calculate inhibition for each independent replicate, summarize variability, and estimate IC50 only when the concentration-response data support a defensible curve fit.

Calculation and blank correction

A commonly used blank-corrected expression is: DPPH scavenging (%) = $[(A_{control} - A_{sample,\ corrected}) / A_{control}] \times 100$.

$A_{sample,\ corrected}$ is the sample-plus-reagent reading after subtracting the sample-only background where that correction is appropriate. Define all blank terms explicitly in the methods section.

IC50 is the concentration associated with 50% scavenging under the stated assay conditions. It should not be extrapolated beyond the tested concentration range without transparent qualification.

Controls and quality assurance

- Reagent/solvent control: establishes the radical signal in the absence of the test material.
- Sample blank: assesses sample colour or turbidity independent of the radical reaction.
- Positive antioxidant reference: verifies that the assay responds to a known comparator.
- Replicates: identify the number and type of replicates and report mean with SD or another justified measure.

- Check time-dependence when comparing samples with different kinetic behaviour; endpoint choice can change apparent ranking.

Interpretation limits

DPPH results are solvent-, pH-, concentration-, reaction-time- and matrix-dependent. They should not be compared across studies unless methods are sufficiently aligned. A DPPH value alone does not justify claims of biological antioxidant efficacy, therapeutic action, or food preservation performance.

References and authoritative resources

1. Brand-Williams W, Cuvelier ME, Berset C. Use of a free radical method to evaluate antioxidant activity. LWT - Food Science and Technology. 1995;28(1):25-30. doi:10.1016/S0023-6438(95)80008-5.

2. Munteanu IG, Apetrei C. Analytical methods used in determining antioxidant activity: A review. International Journal of Molecular Sciences. 2021;22(7):3380. doi:10.3390/ijms22073380.

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